

Topic 1: Design and Analysis of Approximate Arithmetic Units for AI Accelerators

- **Field of Advancement:** Artificial Intelligence (AI) & Low-Power DSP
- **Syllabus Connection:**
 - Module 2: Binary adder – Subtractor.
 - Experiment 1: Design and verification of Adder circuits.

Concept & Motivation: Modern AI neural networks process massive datasets where minor calculation errors are acceptable (e.g., an image pixel value of 253 vs. 255 is indistinguishable to the human eye). "Approximate Computing" is a technique that intentionally simplifies logic circuits to reduce power consumption and chip area, trading off a small amount of accuracy. This is a critical area of research for battery-operated AI devices.

Project Methodology:

1. **Baseline Design:** Design a standard 8-bit Ripple Carry Adder and Carry Lookahead Adder using Verilog HDL (Standard Modules).
2. **Approximate Design:** Design a novel "Approximate Adder" by modifying the logic of the Least Significant Bits (LSBs).
 - *Technique:* Simplify the logic equations for the LSBs (e.g., ignore the carry input for the first 2 bits or use a simple OR gate instead of a Full Adder).
3. **Simulation & Comparison:** Simulate both designs to measure:
 - **Error Rate:** Calculate the Mean Error Distance (MED) between the exact and approximate answers.
 - **Hardware Efficiency:** Synthesize the design to measure Gate Count (Area) and Power Consumption.

Research Output: A comparative study proving that the Approximate Adder saves significant area/power with negligible error, suitable for edge-AI applications.

Approximate Arithmetic Units for AI Accelerators

Current Research Trends (What scientists are studying now):

- **Energy-Efficient Hardware:** Research focuses on "Approximate Computing," a paradigm that trades minor accuracy losses for major gains in speed and battery life. Since AI algorithms (like Neural Networks) are naturally error-tolerant, they do not need perfect math to recognize a cat or a face.
- **Specific Components:** Researchers are currently publishing papers on "Approximate Adders" (like the Lower-Part-OR Adder) and "Approximate Multipliers". They test these

by intentionally removing logic gates from the "Least Significant Bits" (LSBs) to see how much power they can save without breaking the application.

- **Metric of Success:** The standard research goal is to reduce **Area** and **Power Consumption** by 30-50% while keeping the "Mean Error Distance" (MED) low enough for image processing tasks.

Real-World Day-to-Day Applications:

- **Google TPU (Tensor Processing Unit):** Google uses approximate computing principles in its custom AI chips (TPUs) to process search queries and voice recognition faster. They use "Reduced Precision" math (e.g., using 8-bit integers instead of 32-bit floats) to save energy.
 - **Smartphone Cameras:** When your phone processes a photo (HDR, Night Mode), it often uses approximate hardware to process millions of pixels quickly without draining your battery.
 - **Video Streaming:** Services like YouTube or Netflix use approximate algorithms for video compression, where minor pixel errors are allowed because the human eye cannot see them.
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Project Title:

Design and Performance Analysis of Approximate Adders for Energy-Efficient AI Applications

Domain: Digital VLSI Design / Computer Architecture

1. Problem Statement & Motivation: Modern applications like Artificial Intelligence (AI) and Image Processing handle massive amounts of noisy data (e.g., pixels in an image). In these applications, perfect mathematical accuracy is often unnecessary; for example, the human eye cannot distinguish between a pixel value of 254 and 255. Standard arithmetic circuits (like the ones we study in class) differ by focusing on 100% accuracy, which consumes significant power and chip area. There is a growing industry need for "**Approximate Computing**"—hardware that intentionally simplifies logic to gain speed and save battery life.

2. Project Objective: The goal of this project is to design, simulate, and compare two different Adder architectures using **Verilog HDL**:

1. **Exact Adder:** A standard 8-bit Ripple Carry Adder (Base Reference).

2. **Approximate Adder:** A modified adder where the Least Significant Bits (LSBs) are simplified (using logic reduction techniques) to reduce gate count.

3. Connection to Syllabus (BAEEE201): This project directly implements concepts from our **Digital Electronics** course:

- **Module 2 (Combinational Circuits):** We will be designing Binary Adders and analyzing their carry propagation.
- **Module 1 & 5 (HDL & FPGA):** The entire design will be coded in Verilog HDL and synthesized to analyze gate usage (Area) and delay (Speed).

4. Methodology:

- **Step 1:** Code the standard Adder in Verilog.
- **Step 2:** Design the Approximate Adder by modifying the logic equations for the first 2-3 bits (replacing complex Full Adders with simpler OR/XOR gates).
- **Step 3:** Create a **Test Bench** to simulate both designs with 100+ random inputs.
- **Step 4: Research Analysis:** We will generate a comparative table showing the trade-off:
How much Power/Area did we save vs. What was the percentage error introduced?

5. Expected Outcome (Research Component): We aim to publish a research paper titled "*Comparative Analysis of Approximate vs. Exact Adders for Low-Power DSP Applications.*" The project will demonstrate that for specific applications, the Approximate Adder provides a significant hardware advantage (lower cost/power) with negligible impact on accuracy.

6. Tools to be Used:

- Xilinx Vivado / Intel Quartus / ModelSim (for Simulation & Synthesis).